

WORK EXPERIENCE

Loxahatchee River District

Jupiter, FL

IT Intern

May 2025 – Present

- Designed and consumed RESTful APIs to automate ingestion and storage of district call and training data.
- Built automated extraction pipelines for water-distribution data, reducing manual collection time by several hours per week.
- Developed a database-driven tool to auto-generate late-notice letters for customers, improving turnaround time for communications.
- Built a Linux-based network monitoring utility using Kismet to surveil district pump stations.
- Assisted in implementing AI-based image recognition for automated pressure gauge readings.

Kratos Defense & Security Solutions

Jupiter, FL

Digital Engineering Intern

May 2023 – Aug 2023

- Designed functional models in Cameo Systems Modeler to calculate optimal inputs for a turbojet engine.
- Wrote Visual Basic scripts in Excel to calculate expected thrust and cost of a hypothetical engine.
- Collaborated with engineering teams to implement performance calculations for a C++ steady-state engine model.

EDUCATION

University of Florida

Gainesville, FL

B.S. in Computer Science, GPA: 3.66/4.0

Jun 2022 – May 2026

- **Coursework:** Data Structures & Algorithms, Operating Systems, Algorithm Abstraction & Design, Computer Organization, Machine Learning Engineering, Natural Language Processing

PROJECTS

- **Archive Vision:** Developed a mobile application for digitizing and querying archival documents using OCR/HTR pipelines, ChromaDB vector search, and a RAG-based chatbot backed by OpenAI and Gemini. Built on FastAPI with async job processing via Celery and Redis, containerized with Docker Compose. Currently in talks with the University of Florida to expand adoption across university departments.
- **Portfolio Website:** Built a personal portfolio site in React, TypeScript, and Vite featuring interactive simulations including a Boids flocking simulation, Wave Function Collapse, and a Connect Four AI using minimax. Deployed on Cloudflare Pages.
- **“Craven”:** Designed and implemented a recipe-matching web application with a preference-based suggestion algorithm. Built a RESTful backend with Node.js and Express, designed MongoDB schemas for user data, and developed a React frontend consuming backend APIs.
- **3D Graphics Engine:** Designed a 3D graphics engine in C++ using OpenGL and SDL with a fully traversable environment.

SKILLS

Languages: Python, Java, JavaScript, TypeScript, C++, C, Bash

Frameworks: React, Node.js, FastAPI, Express

Tools & Platforms: Docker, Git, Linux, PostgreSQL, MySQL, SQLite, ChromaDB, Redis, Celery, Qt

Concepts: RESTful APIs, Machine Learning, Computer Vision, Vector Search, RAG